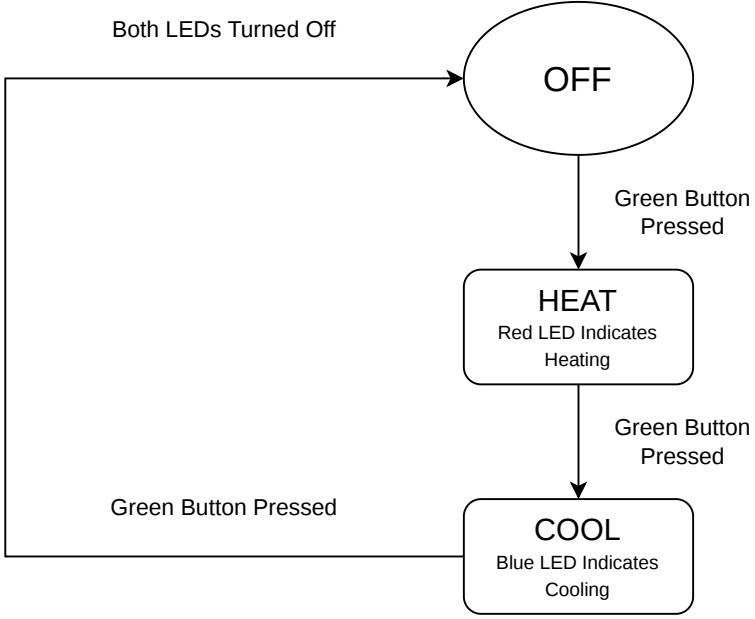

Additional Notes:

- Temperature sensor: Reads room temperature and converts Celsius to Fahrenheit.

- Default set point: 72°F.

- Red button: Increases the set point by 1°F per activation.

- Blue button: Decreases the set point by 1°F per activation.

- Heating behavior: In HEAT mode, the red LED pulses when the measured temperature is below the set point; otherwise, it stays on.

- Cooling behavior: In COOL mode, the blue LED pulses when the measured temperature is above the set point; otherwise, it stays on.

- OFF behavior: Both LEDs are off.

- Display: Shows the current time, temperature, operating mode, and set point.

- Serial output: Reports the current state, temperature, and set point periodically.